

SafeG VMS — Jetson Installation Guide

Complete setup for the on-site Jetson edge device. Assumes the **cloud server is already running**.

Jetson role

Item	Value
Purpose	On-site edge device (kiosk, RFID, local DB)
Local DB	vms
Syncs to	Cloud vms_cloud via database.cloud.* in .env
Code path	/var/www/html/vms
Linux user	tally
URL	http://JETSON_IP/vms/login

1. System packages

```
sudo apt update && sudo apt upgrade -y

sudo apt install -y \
  apache2 libapache2-mod-php \
  php php-cli php-mysql php-mbstring php-xml php-curl php-intl php-gd php-zip php-bcmath \
  mysql-server \
  git unzip composer \
  tesseract-ocr

sudo a2enmod rewrite headers
sudo systemctl enable apache2 mysql
sudo systemctl restart apache2
```

Verify Apache PHP has intl (required)

```
ls /etc/apache2/mods-enabled/ | grep php
```

If you see `php8.2` (common on Jetson/Ubuntu):

```
sudo apt install -y php8.2-intl
sudo systemctl restart apache2
```

Test:

```
echo '<?php echo PHP_VERSION." intl=".(extension_loaded("intl")?"yes":"NO");' | sudo tee
/var/www/html/vms/public/_chk.php
```

```
curl -s http://127.0.0.1/vms/_chk.php
sudo rm /var/www/html/vms/public/_chk.php
```

Must show `intl=yes` . Without intl you get: `Class "Locale" not found` .

2. MySQL (local on Jetson)

```
sudo mysql_secure_installation
sudo mysql -u root -p
```

```
CREATE DATABASE vms CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci;
CREATE USER 'vms_user'@'localhost' IDENTIFIED BY 'YOUR_PASSWORD';
GRANT ALL PRIVILEGES ON vms.* TO 'vms_user'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

3. Clone code (do NOT use sudo)

```
cd /var/www/html
git clone https://github.com/frhanzv/vms.git
cd vms
composer install --no-dev --optimize-autoloader
```

If repo already exists with messy local changes:

```
cd /var/www/html/vms
cp .env ~/.env.backup
git fetch origin main
git reset --hard origin/main
git clean -fd
cp ~/.env.backup .env
composer install --no-dev --optimize-autoloader
```

Fix ownership:

```
sudo chown -R tally:tally /var/www/html/vms
```

If Git says "dubious ownership":

```
sudo chown -R tally:tally /var/www/html/vms
```

4. Configure `.env`

```
cp env .env
nano .env
```

Jetson settings:

```
CI_ENVIRONMENT = production
app.baseURL = 'http://JETSON_IP/vms/'

# Local DB (Jetson)
database.default.hostname = localhost
database.default.database = vms
database.default.username = vms_user
database.default.password = YOUR_PASSWORD
database.default.DBDriver = MySQLi
database.default.port = 3306

# Cloud DB (for sync to server)
database.cloud.hostname = CLOUD_SERVER_IP
database.cloud.database = vms_cloud
database.cloud.username = vms_sync
database.cloud.password = YOUR_CLOUD_SYNC_PASSWORD
database.cloud.DBDriver = MySQLi
database.cloud.port = 3306

encryption.key = hex2bin:GENERATE_NEW_KEY
```

Generate encryption key:

```
php -r "echo 'hex2bin:'.bin2hex(random_bytes(32)).PHP_EOL;"
```

Set mail/LLM vars if needed.

5. Permissions

```
cd /var/www/html/vms

sudo mkdir -p writable/{cache,logs,session,uploads,debugbar}
sudo mkdir -p public/uploads/visitor_photos writable/sync

sudo chown -R tally:www-data writable
sudo chmod -R 775 writable

sudo usermod -aG www-data tally
```

Log out and back in to SSH after `usermod` .

Test write:

```
touch writable/cache/test.txt && rm writable/cache/test.txt && echo "OK"
```

Rule: repo owned by `tally` ; `writable/` owned by `tally:www-data` with `775` .

6. Apache vhost

```
sudo nano /etc/apache2/sites-available/vms.conf
```

```
<VirtualHost *:80>
    ServerName JETSON_IP

    Alias /vms /var/www/html/vms/public
    <Directory /var/www/html/vms/public>
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/vms-error.log
    CustomLog ${APACHE_LOG_DIR}/vms-access.log combined
</VirtualHost>
```

Enable:

```
sudo a2ensite vms.conf
sudo a2dissite 000-default.conf
sudo systemctl reload apache2
```

Keep `RewriteBase /vms` in `public/.htaccess` (default — do not change).

Open: `http://JETSON_IP/vms/login`

7. Migrate & seed

```
cd /var/www/html/vms
php spark migrate
php spark db:seed DatabaseSeeder
```

If `CountriesSeeder` fails (MySQL FK error on `states`), run the rest manually:

```
php spark db:seed VisitorTypeSeeder
php spark db:seed WorkflowSeeder
php spark db:seed ClientFeaturesSeeder
php spark db:seed ClientFormFieldsSeeder
php spark db:seed ClientNotificationSettingsSeeder
```

If cache permission error:

```
sudo -u www-data php spark migrate
```

Demo login (change on production)

Username	Password	Role
admin	admin123	superadmin
host	host123	host
officer	officer123	officer
approver	approver123	admin

8. Sync with cloud (first time)

Ensure cloud allows Jetson to reach MySQL on port 3306 (cloud firewall + MySQL user `vms_sync`).

On Jetson:

```
cd /var/www/html/vms
php spark sync:run --full
```

Normal sync after that:

```
php spark sync:run
```

Optional cron (every 5 minutes):

```
crontab -e
```

```
* /5 * * * * cd /var/www/html/vms && php spark sync:run --quiet >>
/var/www/html/vms/writable/logs/sync-cron.log 2>&1
```

9. Verification checklist

Check	How
Login page	<code>http://JETSON_IP/vms/login</code>
Demo login	<code>admin / admin123</code>
Migrations	<code>php spark migrate:status</code>
Apache intl	<code>curl test shows intl=yes</code>
Writable OK	<code>touch writable/cache/test works</code>
Logs	<code>tail writable/logs/log-*.log</code>

Sync	php spark sync:run
------	--------------------

10. After every git pull on Jetson

```
cd /var/www/html/vms
cp .env ~/.env.backup
git pull origin main
cp ~/.env.backup .env

composer install --no-dev --optimize-autoloader
php spark migrate

sudo chown -R tally:www-data writable
sudo chmod -R 775 writable
rm -rf writable/cache/*
sudo systemctl restart apache2
```

If pull blocked by local changes:

```
cp .env ~/.env.backup
git fetch origin main
git reset --hard origin/main
git clean -fd
cp ~/.env.backup .env
```

Never use `sudo git pull .`

Common Jetson errors

Error	Fix
Cache unable to write to writable/cache/	<code>sudo chown -R tally:www-data writable && sudo chmod -R 775 writable</code>
Class "Locale" not found	<code>sudo apt install php8.2-intl</code> (match Apache PHP version)
Session permission denied	Same writable fix above
git dubious ownership	<code>sudo chown -R tally:tally /var/www/html/vms</code>
git pull blocked	Backup .env, then <code>git reset --hard origin/main</code>
Broken URLs / redirects	<code>app.baseURL = 'http://JETSON_IP/vms/' + keep RewriteBase /vms</code>

Quick order (fresh Jetson install)

1. Install packages (+ **php-intl** for Apache's PHP version)
2. Create local MySQL `vms` + `vms_user`
3. Clone repo + `composer install`

4. Configure `.env` (local DB + cloud sync DB)
5. Set permissions (`tally:www-data` on `writable/`)
6. Apache vhost with `Alias /vms`
7. `php spark migrate + seed`
8. `php spark sync:run --full`
9. Open `http://JETSON_IP/vms/login`